ECE2150J - RC2 - Zhao Guojiao

https://github.com/CatGPT-Zoe/ECE2150J-SJTU-Global-College

More Exercises

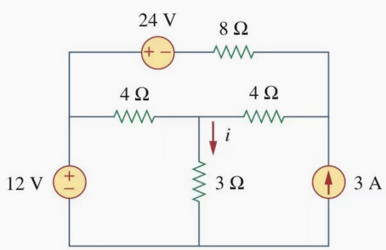
1 Superposition

Steps

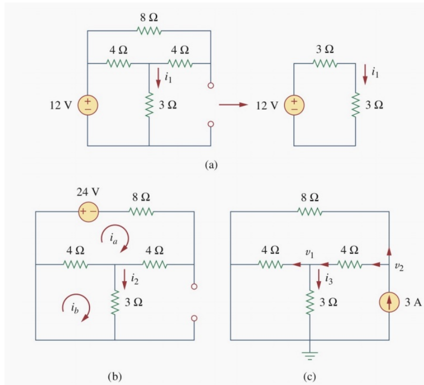
1. Only consider one **independent** source.
   - voltage source: short circuit
   - current source: open circuit
2. Use additivity.

Example :

Find i in the circuit.



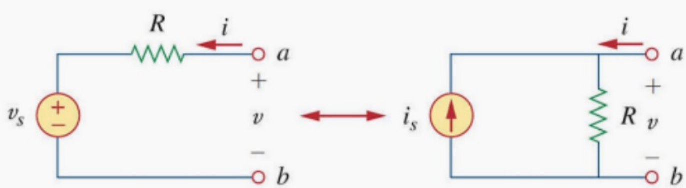
Answer: 3A



2 Source Transformations

We can replace a voltage source with a resistance with a corresponding current source with the same resistance to simplify the circuit.

In the case shown below, $v_s = i_s \times R$



(a) (b)

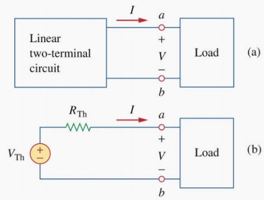
direction!

For dependent sources, the source transformation is also valid.

3 Thevenin's theorem

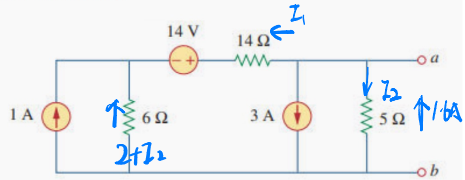
Thevenin's Theorem

A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}.



- V_{Th}: the open-circuit voltage at the terminals.
- R_{Th}: the equivalent resistance at the terminals when all the independent sources are turned off.

Obtain the Thevenin equivalent circuit of this circuit with respect to terminal a and b.



Answer: $V_{Th} =$ - 8 V
$R_{Th} =$ 4Ω

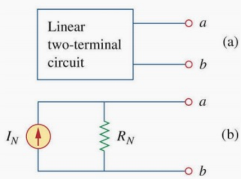
pay attention to the sign of the voltage !

$$\begin{cases} I_1+I_2+3=0 \\ +14I_1-5I_2-6(2+I_2)+14=0 \end{cases}$$

$I_1=-1.4A$ $I_2=-1.6A$

4 Norton's theorem

A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_{Th} in parallel with a resistor R_{Th}.



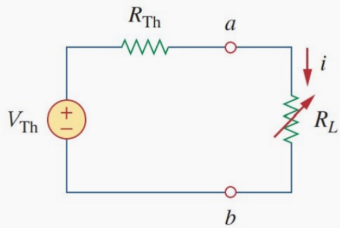
- I_{Th}: the short-circuit current at the terminals.
- R_{Th}: the equivalent resistance at the terminals when all the independent sources are turned off.

5 Maximum power transfer

A circuit is usually designed to provide power to a load. For different kinds of circuits, we have different concerns

- **Maximum Power Efficiency:** In power utility systems, the amount of electricity is very large. Therefore, how to increase the efficiency of power transfer becomes an important problem.
- **Maximum Power Transfer:** In communication and instrumental systems, the amount of electricity is small so the problem of efficiency is not so important. Instead, we want to transfer as much of power as possible to the load.

The Thevenin's equivalent circuit is useful in finding the maximum power delivered to a load. In the circuit below, R_L represents the load.



Since

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th}+R_L}\right)^2 R_L$$

Let $\frac{dP}{dR_L} = V_{Th}^2 \frac{R_{Th}-R_L}{(R_{Th}+R_L)^3} = 0$, we have $R_L = R_{Th}$.

And when $R_L = R_{Th}$, $\frac{d^2P}{dR_L^2} = V_{Th}^2 \frac{2R_l-4R_{Th}}{(R_{Th}+R_L)^4} = -\frac{V_{Th}^2}{8R_{Th}^2} < 0$.

Thus p reaches maximum at $R_L = R_{Th}$. $p_{max} = \frac{V_{Th}^2}{4R_{Th}}$

6 Ideal Op-amp

Characteristics of ideal op-amp:

- Open circuit at two input terminals ($i_1 = i_2 = 0$)
- Same voltage at two input terminals ($v_1 = v_2$)
- Do not apply KCL to the output terminals!

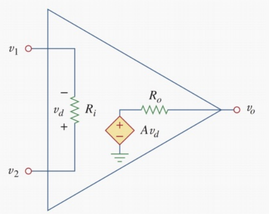


Figure: Op-amp's equivalent circuit

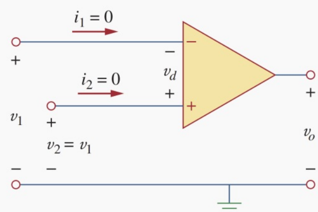
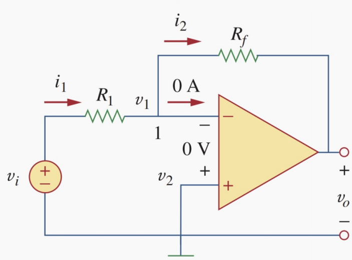


Figure: Symbol of ideal op-amp

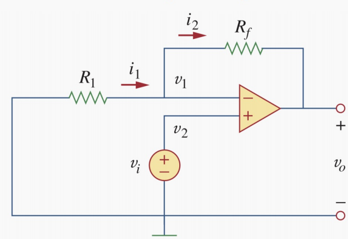
Inverting Amplifier

$$A_v = \frac{v_o}{v_i} = -\frac{R_f}{R_1}$$



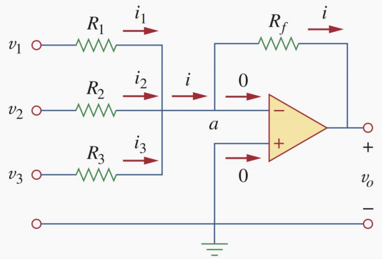
Non-inverting Amplifier

$$A_v = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_1}$$



Summing Amplifier

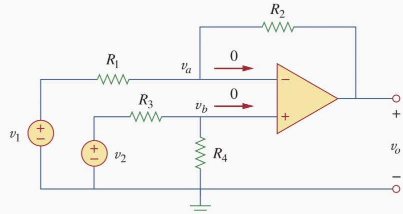
$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$



Difference Amplifier

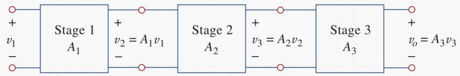
Check if $R_1 = R_2,\ R_3 = R_4$

$$v_o = \left(\frac{R_2}{R_1}+1\right)\frac{R_4}{R_3+R_4}v_2 - \frac{R_2}{R_1}v_1$$



Cascaded Op Amps

Gain of Cascaded Op Amp



Original input signal is increased by the gain of the individual stage, and the final gain is the **product of all gains at each stage**.

$$A = \frac{v_o}{v_1} = \frac{v_2}{v_1}\cdot\frac{v_3}{v_2}\cdot\frac{v_o}{v_3} = A_1A_2A_3$$

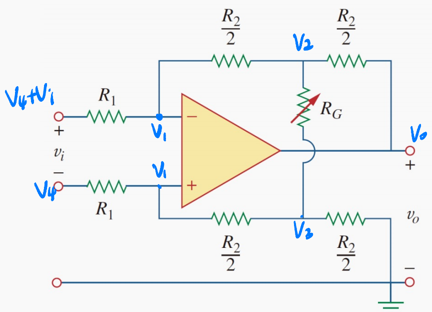
For basic op-amp circuits:

Op-amp circuits	Input-output relationship
Inverting amplifier	$A = \frac{v_0}{v_i} = -\frac{R_f}{R_1}$
Non-inverting amplifier	$A = \frac{v_0}{v_i} = 1 + \frac{R_f}{R_1}$
Summing amplifier	$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$
Difference amplifier	$v_o = \left[\left(\frac{R_2}{R_1}+1\right)\left(\frac{R_4/R_3}{1+R_4/R_3}\right)\right] v_2 - \left[\frac{R_2}{R_1}\right] v_1$

7 Exercises

1 Show that:

$$\frac{v_o}{v_i} = \frac{-R_2}{R_1}\left(1+\frac{R_2}{2R_G}\right)$$



$$\begin{cases} \dfrac{V_1-(V_4+V_i)}{R_1} + \dfrac{V_1-V_2}{\frac{R_2}{2}} = 0 & ① \quad (i_1=0) \\ \dfrac{V_1-V_4}{R_1} + \dfrac{V_1-V_3}{\frac{R_2}{2}} = 0 & ② \quad (i_2=0) \end{cases}$$

$$①-②: \quad -\frac{V_i}{R_1} - \frac{2(V_2-V_3)}{R_2} = 0$$

$$\Rightarrow V_i = -\frac{2R_1}{R_2}(V_2-V_3)$$

$$\begin{cases} \dfrac{V_2-V_1}{\frac{R_2}{2}} + \dfrac{V_2-V_3}{R_G} + \dfrac{V_2-V_o}{\frac{R_2}{2}} = 0 \\ \dfrac{V_3-V_1}{\frac{R_2}{2}} + \dfrac{V_3-V_2}{R_G} + \dfrac{V_3}{\frac{R_2}{2}} = 0 \end{cases}$$

$$\frac{2V_2}{R_2} + \frac{V_2-V_3}{R_G} + \frac{2(V_2-V_o)}{R_2} = \frac{2V_3}{R_2} + \frac{V_3-V_2}{R_G} + \frac{2V_3}{R_2}$$

$$\Rightarrow V_o = \left(2+\frac{R_2}{R_G}\right)(V_2-V_3)$$

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1}\left(1+\frac{R_2}{2R_G}\right)$$

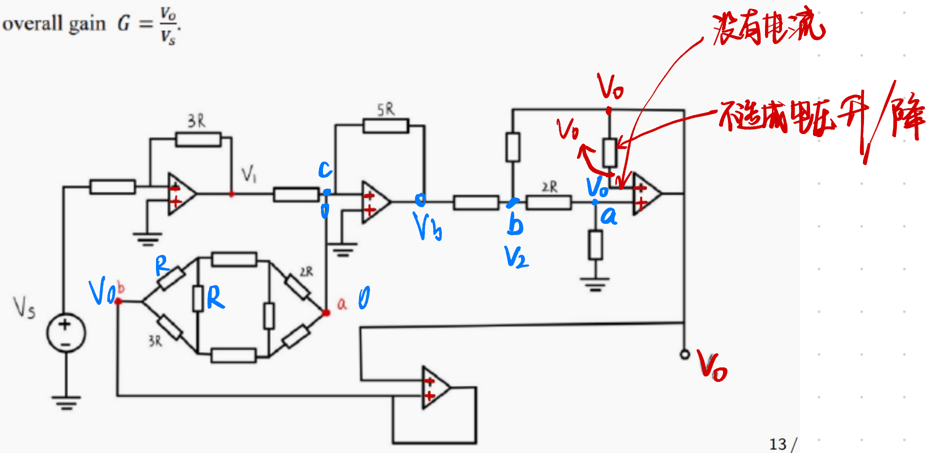
②

4. Suppose all resistors without labels have resistance of R. Please answer the following questions. [16 points]

(1) V_1 in terms of V_s and R.

(2) The equivalent resistance R_{ab} between terminals a and b.

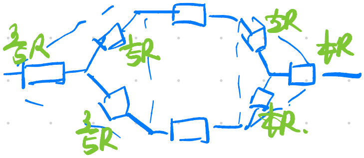
(3) The overall gain $G = \frac{V_o}{V_s}$.



(1) verting Amp

$$V_1 = -\frac{3R}{R}V_s = -3V_s$$

(2)



$$R_{ab} = \frac{3}{5}R + \left(\frac{1}{5}R + R + \frac{2}{5}R\right) \parallel \left(\frac{3}{5}R + R + \frac{4}{5}R\right) + \frac{2}{5}R$$

$$= \frac{141}{71}R = 1.986R$$

(3)

node a
node b
node c

$$\begin{cases} \dfrac{V_0 - V_2}{2R} + \dfrac{V_2}{R} = 0 \\[2mm] \dfrac{V_2 - V_3}{R} + \dfrac{V_2 - V_0}{R} + \dfrac{V_2}{2R + 4R} = 0 \\[2mm] \dfrac{3V_s}{R} - \dfrac{V_3}{5R} - \dfrac{V_0}{\frac{141}{71}R} = 0 \end{cases}$$

$$\Rightarrow \quad V_3 = 6V_0$$

$$3V_s - \frac{6}{5}V_0 - \frac{71}{141}V_0 = 0$$

$$\frac{V_0}{V_s} = \frac{2115}{1201} = 1.76$$